



Education

Degree	Institute	CGPA	Years
PhD	Indian Institute of Technology Madras	9.5	2022-ongoing
M.Tech	Indian Institute of Technology Jodhpur	9.15	2020 - 22
B. Tech	Veer Surendra Sai University of Technology, Sambalpur	9.18	2016 - 20

Projects

M.Tech Project: Role of microstructure and bio-corrosion behavior of Mg rich Mg-Zn-Nd alloys

The work incorporated neodymium additions (upto 5%) in Mg-4Zn alloy in order to improve mechanical and corrosion resistance property which found its application in biomedical implants.

B.Tech Project: Effect of Sintering Temperature and Reinforcement Composition on Microstructure and Mechanical properties of Al-SiC-Fly Ash Hybrid Composite using Powder Metallurgy Route

In the present study, Al-15SiC-5FA and Al-12.5SiC-7.5FA hybrid composites are fabricated using powder metallurgy (P/M) route by varying the sintering temperature followed by microstructural analysis and mechanical properties evaluation that includes hardness, density, and compressive strength (CS).

Achievements

- Poster presentation in the 7th **International Conference** on Processing and Characterization of Materials (ICPCM-2025), NIT Rourkela, 12nd -13th December, 2025.
- Poster presentation in the 30th **International Conference** on Processing of Advanced Materials and Fabrication of Products (PAMFP-2024), IIT Hyderabad, 11st -13th December, 2024.
- Poster presentation in the 28th **International Conference** on Non-ferrous Metals - 2024 (ICNFM - 2024), Bhubaneswar, 12nd -13th July, 2024.
- Poster presentation in the **National Symposium** - "Recent Trends in Material Science", AMALGAM, IIT MADRAS, 3rd March, 2024.
- Paper presentation in the "2nd Virtual **National Conference** on Advancement in Materials Processing Technology" at NIT, Jamshedpur, 31st October–2nd November, 2020.
- Paper presentation in "Webinar for Young Researchers in Metallurgical & Materials Science", CSIR – National Metallurgical Laboratory, Jamshedpur, 12th – 14th August, 2020.
- Paper presentation in "**International Conference** on Innovations in Aluminium Technology" at IIM, Angul, 8th-9th February, 2020.
- Paper presentation in the "4th **International Conference** on Advances in Steel, Power and Construction Technology" at OPJindal University, Raigarh during 21st -22nd November, 2019.
- Poster presentation entitled in TEQUIP-III sponsored "NCFAM - **National Conference** on Frontiers in Advanced Materials" at VSSUT, Burla, Odisha during 27th - 28th July, 2019.

Industrial Training and Internships

- ThermoCalc training, organised by **Bhanu Scientific Systems Pvt Ltd (BSSPL)**, Hyderabad (3rd - 6th May, 2023).
- Summer Research Training Programme, **CSIR-Institute of Minerals and Materials Technology**, Bhubaneswar (1st June – 31st August, 2020).
- Vocational training in Cast House, Smelter Plant, **Aditya Aluminium and Power Plant**, Lapanga, Sambalpur (3rd June – 3rd July, 2019).
- Research intern at **National Institute of Technology**, Rourkela (15th December–31st December, 2018).
- Vocational training in Blast Furnace, **SAIL, Rourkela Steel Plant** (28th May – 27th June, 2018).

Skills and Expertise

Metallography: Expertise on the metallography of ferrous and non-ferrous alloys.

Mechanical testing: Tensile testing, and various aspects of hardness testing.

Characterization: Optical microscopy, SEM, TEM, HRTEM, XRD, ImageJ.

Software skills: Thermo-calc, Phase field simulation, Origin Pro, C Programming and C++, Microsoft Office.

Awards

- **2nd position** in Poster presentation at International Conference on Processing of Advanced Materials and Fabrication of Products, 2024, IIT Hyderabad.
- **1st position** in Poster presentation at International Conference on Non-ferrous Metals (ICNFM), 2024, Bhubaneswar.
- **1st position** in Sri Vasudevan Quiz (Meta Quiz) at Amalgam, Department of Metallurgical and Materials Engineering, IIT Madras, 2023.
- **Silver medal** for the Best Academic Performance of the class of M.Tech Materials Engineering Program, IIT Jodhpur, 2022.
- **10th position** in the META EXPO, STUDYMETA, 2022.
- **1st position** in Online Technical Presentation at Diamond Jubilee Year Celebration, NIT Rourkela, 2021.
- **1st position** in Online Quiz Contest at Diamond Jubilee Year Celebration, NIT Rourkela, 2021.
- **Special Prize** in Doodling the Expression on the Social Fest, IIT Jammu, 2021.
- **2nd position** in Poster presentation at IIM, Reminiscence organised by IEST, Shibpur.
- **2nd position** in slogan poster competition on the event of International day of Yoga.
- **1st position** in quiz on the event of International day of Yoga.

Declaration

I hereby declare that the information stated above is true to the best of my knowledge.

Dt – 10/04/2026

Arpita Priyadarshani Samal